



Specifying and Analyzing Transactional Consistency Models with Predicates

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A semantic and graph-theoretic foundation for predicate operations

Item operations vs. predicate operations

Read(x, v)

Read(2, 60)

Write(x, v)

Write(3, 40)

point access
one value

Employee

EID	Salary
0	80
1	100
2	60
3	40

PredRead(P, M)

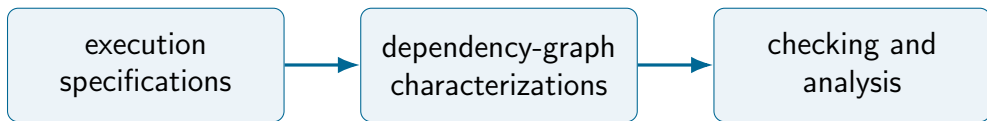
$P : \text{Salary} < 80$

condition-based access
a set of values

One query can observe both matching and non-matching employees.

Review: the item-only foundation

What we know about transactional consistency
without predicates



The classical correspondence is built around named items and item dependencies.

Review: specifications for item operations

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
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Serializability

equivalent to a serial execution

A Critique of ANSI SQL Isolation Levels

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Snapshot isolation

a consistent snapshot + write-conflict
control

Eswaran et al., 1976; Berenson et al., 1995

Review: axiomatic specifications

A shared execution vocabulary: (VIS, AR)

A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

atomic visibility

Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks
The University of Texas at Austin and Cornell University

Lorenzo Alvisi
The University of Texas at Austin and Cornell University

client-centric views

Yuer Pu
Cornell University

Allen Clement
Google, Inc.

On the Complexity of Checking Transactional Consistency

RANADEEP BISWAS, Université de Paris, IRIF, CNRS, France
CONSTANTIN ENEA, Université de Paris, IRIF, CNRS, France

checking complexity

Visibility says which item versions a transaction may observe; **arbitration** orders writes.

Cerone, Bernardi, and Gotsman, 2015; Crooks et al., 2017

Review: SER has an exact graph test

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CONCURRENCY CONTROL FOR DATABASE SYSTEMS

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EXTENDED ABSTRACT

Item dependency graph

$$\mathcal{H} \models \text{SER}$$


internally consistent +
acyclic graph

Eswaran et al., 1976; Bernstein, Hadzilacos, and Goodman, 1987

Review: SI has a cycle condition

4.2 Snapshot Isolation

A transaction executing with *Snapshot Isolation* always reads data from a snapshot of the (committed) data as of the time the transaction started, called its *Start-Timestamp*. This time may be any time before the transaction's first Read. A transaction running in Snapshot Isolation is never blocked attempting a read as long as the snapshot data from its *Start-Timestamp* can be maintained. The transaction's writes (updates, inserts, and deletes) will also be reflected in this snapshot, to be read again if the transaction accesses (i.e., reads or updates) the data a second time. Updates by other transactions active after the transaction *Start-Timestamp* are invisible to the transaction.

Snapshot Isolation is a type of multiversion concurrency control. It extends the Multiversion Mixed Method described in [BHG], which allowed snapshot reads by read-only transactions.

When the transaction T_1 is ready to commit, it gets a *Commit-Timestamp*, which is larger than any existing *Start-Timestamp* or *Commit-Timestamp*. The transaction successfully commits only if no other transaction T_2 with a *Commit-Timestamp* in T_1 's interval [*Start-Timestamp*, *Commit-Timestamp*] wrote data that T_1 also wrote. Otherwise, T_1 will abort. This feature, called *First-committer-wins* prevents lost updates (phenomenon P4). When T_1 commits, its changes become visible to all transactions whose *Start-Timestamps* are larger than T_1 's *Commit-Timestamp*.

SI dependency graph

$$\mathcal{H} \models \text{SI}$$

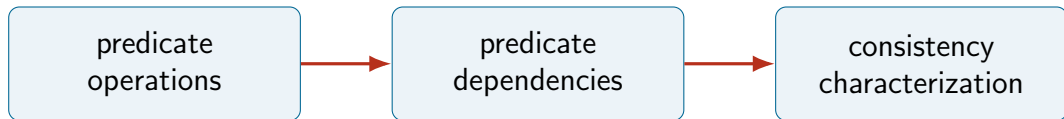


internally consistent +
every cycle contains **two adjacent**
anti-dependency edges

These results explain item histories. What changes when a read is a predicate?

Review: adding predicates changes the object of analysis

What we know about transactional consistency
with predicates



The question is no longer only which version of an item was read.

Review: specifications still speak in item-level terms

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SER

serial execution

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SI

snapshot + write-conflict control

Predicate semantics is not supplied by these item-level specifications.

Eswaran et al., 1976; Berenson et al., 1995

Review: predicate graphs extend serializability analysis

Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

Generalized Isolation Level Definitions

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Predicate dependency graph

Predicate reads induce
additional dependency edges

But which later writers should be
connected?

Adya, 1999; "Generalized Isolation Level Definitions" 2000

Review: predicate anti-dependency is a design choice

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

earliest match-changing
writer

PredRW
any match-
changing writer

On the Complexity of Checking Mixed Isolation
Levels for SQL Transactions

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every later writer

$$\text{PredRW}^- \subseteq \text{PredRW} \subseteq \text{PredRW}^+$$

Different edge sets; potentially different checking costs.

Sun and Zou, 2025; Bouajjani, Enea, and Román-Calvo, 2025

Review: SI inherits the same predicate-edge question

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SI cycle test

Every cycle needs two adjacent
anti-dependencies

Do predicate anti-dependencies preserve this
characterization?

Adya, 1999; Cerone and Gotsman, 2018

Review: existing axioms do not model predicate operations

A Framework for Transactional Consistency Models with Atomic Visibility

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(VIS, AR) execution model

Seeing is Believing: A Client-Centric Specification of Database Isolation

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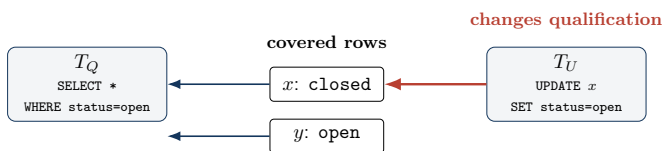
Allen Clement
Google, Inc.

client-centric specifications

Missing: an explanation of both returned and omitted predicate items.

Cerone, Bernardi, and Gotsman, 2015; Crooks et al., 2017

A predicate result contains more than returned rows



result: $\{y\}$

Key observation: $x \notin \text{result}$
still records an observation of x .

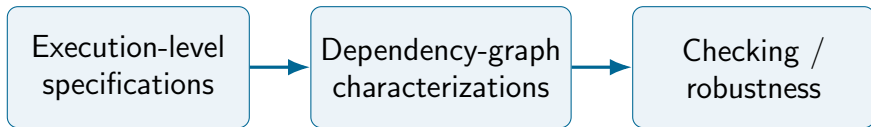
Item reads: value observed

Predicate reads: returned rows and excluded rows

Phantom-style interactions need an explanation for both.

Berenson et al., 1995; Adya, 1999

The gap: familiar foundations stop at named items



Predicate operations are missing from the semantic-graph correspondence.

Question: Can we recover an exact, predicate-aware foundation for SER and SI?

Semantic anchor: explain every covered item

VIS/AR execution framework

Visible transactions

provide the version set



Latest visible writer

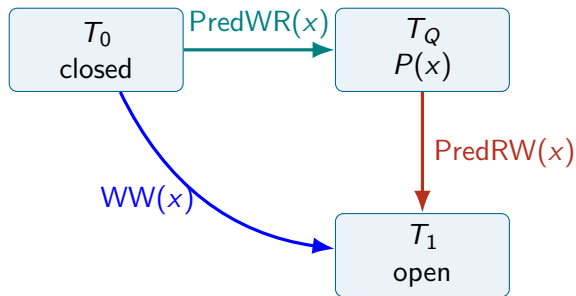
explains each covered item

Returned item: witness matches P

Omitted item: witness does not match P

Cerone, Bernardi, and Gotsman, 2015

From a semantic witness to graph edges



PredWR: version witness

PredRW: later writer changes the observed result

Result-sensitive: $\Delta(T_0, T_Q, T_1, x)$

The graph records why the result was observed — including the absence of x .

Exact graph characterizations

Serializability

$$\mathcal{H} \models \text{SER}$$



internally consistent +
acyclic predicate-aware graph

Snapshot isolation

$$\mathcal{H} \models \text{SI}$$



internally consistent +
every cycle has **two adjacent**
RW / PredRW edges

Not analogy: both equivalences are proved against the predicate-aware axioms.

Cerone and Gotsman, 2018

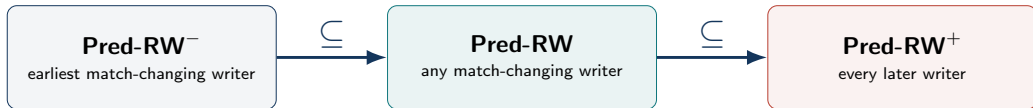
Why the proof bridge matters



observable histories $\longleftrightarrow$ executions

- Predicate graphs are **observable-history** objects.
- Axiomatics supplies the **semantic** correctness target.
- The bridge validates graph-based analysis beyond item-only workloads.

Related work disagrees on one crucial edge



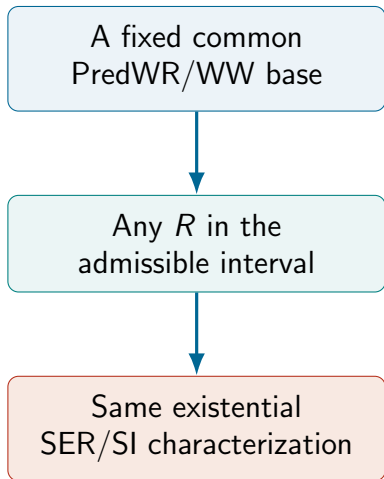
Same existential history-level characterization of SER and SI
edge sets and checking cost may differ

Sun and Zou, 2025

Adya, 1999; “Generalized Isolation
Level Definitions” 2000

Bouajjani, Enea, and Román-Calvo,
2025

What the design space tells us



The theorem is semantic, not a claim that edge sets coincide.

Practical consequence: choose an admissible edge policy for an analysis, while retaining its history-level meaning.

Checking cost can still differ.

Future work: turn the foundation into analysis tools

Efficient black-box
checking

Recover version sets
and version order

Robustness analysis

Extend static reasoning
to predicate operations

Both need a semantics-preserving predicate dependency graph.

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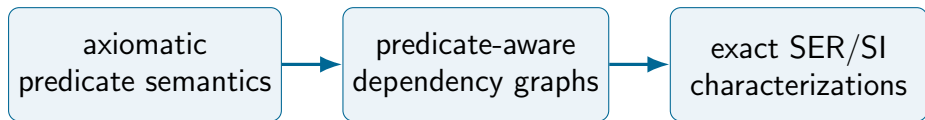
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Predicate results are observations over a covered set.



A principled basis for checking and analyzing databases with predicates.